

A Short Guide to Lemma C.3

The residual subinstrument of a covariant hybrid map

Abstract

Lemma C.3 describes every exactly covariant hybrid map as a convolution of the translated input with one residual quantum subinstrument. Informally, the deterministic part of the classical output is $\sqrt{\gamma}$ times the classical input, while a residual variable records all additional classical error and its correlated quantum operation. This note explains measure-valued instruments, the approximate-identity construction of the residual object, the proof of the convolution formula, and the specialization to the Gaussian input used in Proposition C.4.

1 The structural question

Let

$$\mathfrak{T} = L^1(\mathbf{H}; \mathcal{T}_1(\mathcal{F}_{\text{orb}})), \quad \mathfrak{A} = C_0(\mathbf{H}; \mathcal{K}(\mathcal{F}_{\text{orb}})), \quad \mathfrak{M} = \mathfrak{A}^*.$$

An element $R \in \mathfrak{T}$ is a classical–quantum input: at the classical value $x \in \mathbf{H}$, it carries the trace-class operator $R(x)$. A positive element of $\mathfrak{M}$ is an operator-valued Radon measure on the output classical space.

Suppose

$$\Gamma : \mathfrak{T} \longrightarrow \mathfrak{M}$$

is completely positive, trace nonincreasing, and covariant under both classical translations and oscillator displacements. Classical covariance has gain $\sqrt{\gamma}$: translating the input by x translates the output by $\sqrt{\gamma}x$. The question is whether the part of the output that remains after removing this forced translation can be described by one object independent of x . Lemma C.3 answers yes.

2 Background: quantum instruments and subinstruments

A quantum instrument assigns to every Borel event $E \subset \mathbf{H}$ a completely positive map $\mathcal{I}_E$ on trace-class operators. For an input state X , the number

$$\text{Tr } \mathcal{I}_E(X)$$

is the probability that the classical outcome lies in E , and the positive operator $\mathcal{I}_E(X)$ is the corresponding unnormalized posterior quantum state. Countable additivity in E is required in trace norm.

A *subinstrument* permits total probability below one:

$$\text{Tr } \mathcal{I}_{\mathbf{H}}(X) \leq \text{Tr } X \quad (X \geq 0).$$

This is appropriate here because the weak-* limiting map obtained in Lemma C.2 may lose mass at infinity. It is convenient to package all branches as a measure-valued completely positive map

$$\mathcal{I} : \mathcal{T}_1(\mathcal{F}_{\text{orb}}) \longrightarrow C_0(\mathbf{H}; \mathcal{K}(\mathcal{F}_{\text{orb}}))^*.$$

The value $\mathcal{I}(X)(E)$ is exactly $\mathcal{I}_E(X)$.

3 The expected convolution formula

Imagine first that the input is concentrated at one classical point x and has quantum part X . Translation covariance says that its output should be a translate by $\sqrt{\gamma}x$ of the output obtained at $x = 0$. If $\mathcal{I}(X)$ denotes this origin output, then

$$\Gamma(\delta_x X)(E) = \mathcal{I}(X)(E - \sqrt{\gamma}x).$$

An L^1 input is a continuous mixture of such point inputs, so one expects

$$[\Gamma(R)](E) = \int_{\mathbf{H}} [\mathcal{I}(R(x))](E - \sqrt{\gamma}x) dx. \quad (1)$$

There is one technical obstruction: $\delta_x X$ is not an element of $L^1(\mathbf{H}; \mathcal{T}_1(\mathcal{F}_{\text{orb}}))$. Therefore $\Gamma(\delta_0 X)$ is not directly defined. The proof replaces the point mass by shrinking L^1 bumps and extracts a weak-* limit.

4 Statement of Lemma C.3

Lemma C.3 (rephrased). *Let $\Gamma : \mathfrak{T} \rightarrow \mathfrak{M}$ be completely positive, trace nonincreasing, and exactly covariant under the hybrid translation–Weyl group. There exists a completely positive, trace-nonincreasing measure-valued map*

$$\mathcal{I} : \mathcal{T}_1(\mathcal{F}_{\text{orb}}) \longrightarrow C_0(\mathbf{H}; \mathcal{K}(\mathcal{F}_{\text{orb}}))^*$$

whose Borel branches $\mathcal{I}_E$ form a subinstrument and such that

$$[\Gamma(R)](E) = \int_{\mathbf{H}} [\mathcal{I}(R(x))](E - \sqrt{\gamma}x) dx. \quad (2)$$

Each branch is displacement covariant with amplitude gain $\sqrt{\gamma}$. If Γ is covariant under independent oscillator phase rotations, then every branch is phase covariant as well.

For the input $R(x) = \varphi_{0, \Sigma_p}(x) \Phi_0$, if $\Xi = \Gamma(R)$, then for bounded measurable a and bounded oscillator observable M ,

$$\int a(y) \text{Tr}[M \Xi(dy)] = \int_{\mathbf{H}} \left(\int_{\mathbf{H}} a(y) \varphi_{0, \gamma \Sigma_p}(y - r) dy \right) \text{Tr}[M \mathcal{I}(\Phi_0)(dr)]. \quad (3)$$

Equation (2) is an equality of trace-class-valued Radon measures. Equation (3) is its weak form against a classical test function and a quantum observable.

5 Proof, step by step

5.1 Step 1: approximate an input at the origin

Choose nonnegative functions $u_j \in C_c(\mathbf{H})$ such that

$$\int_{\mathbf{H}} u_j(x) dx = 1, \quad \text{supp } u_j \downarrow \{0\}.$$

For $X \in \mathcal{T}_1(\mathcal{F}_{\text{orb}})$, define

$$\mathcal{I}_j(X) = \Gamma(u_j X), \quad (u_j X)(x) = u_j(x) X. \quad (4)$$

The field $u_j X$ is a normalized bump around the origin with quantum state X . Each $\mathcal{I}_j$ is completely positive. For $X \geq 0$,

$$\mathrm{Tr} \mathcal{I}_j(X) \leq \|u_j X\|_1 = \mathrm{Tr} X,$$

so it is trace nonincreasing.

The sequence need not converge in norm. Pass to adjoints

$$\mathcal{I}_j^* : C_0(\mathbf{H}; \mathcal{K}(\mathcal{F}_{\mathrm{orb}})) \longrightarrow \mathcal{B}(\mathcal{F}_{\mathrm{orb}}).$$

They are completely positive contractions. Separability and a diagonal weak-* extraction on a countable dense operator system give a pointwise weak-* limit $\mathcal{I}^*$. Because every matrix positive cone is weak-* closed, $\mathcal{I}^*$ is completely positive. Define $\mathcal{I}$ by duality.

If (e_ℓ) is a contractive approximate identity of the test algebra, then for $X \geq 0$,

$$\mathrm{Tr} \mathcal{I}(X) = \sup_{\ell} \langle X, \mathcal{I}^*(e_\ell) \rangle \leq \mathrm{Tr} X.$$

Thus the limit is still trace nonincreasing. This is the residual measure-valued map.

5.2 Step 2: prove the convolution identity on simple tensors

First take

$$R(x) = f(x)X, \quad f \in L^1(\mathbf{H}), \quad X \in \mathcal{T}_1(\mathcal{F}_{\mathrm{orb}}).$$

Let

$$(\mathrm{sh}_x u_j)(v) = u_j(v - x).$$

Since

$$(f * u_j)(v)X = \int_{\mathbf{H}} f(x)(\mathrm{sh}_x u_j)(v)X \, dx,$$

bounded linearity, Fubini's theorem, and classical translation covariance give

$$\Gamma((f * u_j)X) = \int_{\mathbf{H}} f(x) \, \mathrm{sh}_{\sqrt{\gamma}x} \mathcal{I}_j(X) \, dx, \tag{5}$$

where

$$(\mathrm{sh}_s \Xi)(E) = \Xi(E - s).$$

The left side converges in measure norm because $f * u_j \rightarrow f$ in L^1 and Γ is bounded. To pass to the limit on the right, pair with any $A \in C_0(\mathbf{H}; \mathcal{K}(\mathcal{F}_{\mathrm{orb}}))$. For each x , weak-* convergence of $\mathcal{I}_j(X)$ gives convergence of the pairing. The integrand is bounded in absolute value by

$$|f(x)| \|X\|_1 \|A\|,$$

which is integrable. Dominated convergence therefore changes $\mathcal{I}_j$ to $\mathcal{I}$ in (5). Uniqueness of the representing Radon measure gives

$$[\Gamma(fX)](E) = \int f(x) [\mathcal{I}(X)](E - \sqrt{\gamma}x) \, dx.$$

Finite sums of simple tensors are dense in $L^1(\mathbf{H}; \mathcal{T}_1(\mathcal{F}_{\mathrm{orb}}))$. Both sides are bounded by $\|R\|_1$, so the formula extends by continuity to every R , proving (2).

5.3 Step 3: verify that the Borel branches form a subinstrument

Complete positivity of the measure-valued map means that, whenever $[X_{kl}] \geq 0$, the matrix

$$[\mathcal{I}(X_{kl})]$$

is a positive matrix-valued measure. Approximate the indicator of a Borel set E by nonnegative continuous functions. Evaluation on E preserves matrix positivity, hence

$$[\mathcal{I}_E(X_{kl})] \geq 0.$$

Therefore every branch $\mathcal{I}_E$ is completely positive.

For $X \geq 0$, positivity of the measure implies

$$0 \leq \mathcal{I}_E(X) \leq \mathcal{I}_H(X),$$

and consequently

$$\text{Tr } \mathcal{I}_E(X) \leq \text{Tr } X.$$

Thus every branch is trace nonincreasing and bounded. For disjoint events, countable additivity holds in trace norm on positive inputs because the increments are positive and their traces add. Decomposing a general trace-class operator into linear combinations of positive ones extends the claim to all inputs.

5.4 Step 4: pass quantum covariance to every branch

Let $D(w)$ be a Weyl displacement. The full covariance of Γ implies, already for the approximants,

$$\mathcal{I}_j(D(w)XD(w)^*) = D(\sqrt{\gamma}w)\mathcal{I}_j(X)D(\sqrt{\gamma}w)^*.$$

Here the equality is one of operator-valued measures; the classical event is not moved by a purely quantum displacement. Passing to the weak-* limit and evaluating on E gives

$$\mathcal{I}_E(D(w)XD(w)^*) = D(\sqrt{\gamma}w)\mathcal{I}_E(X)D(\sqrt{\gamma}w)^*.$$

The same argument applies to independent phase rotations. This branchwise covariance is essential: Proposition C.4 applies a quantum stochastic-order bound separately to every residual event E .

5.5 Step 5: specialize to the Gaussian input

Set

$$R(x) = \varphi_{0, \Sigma_p}(x)\Phi_0.$$

The convolution formula says that one may generate the output by the following formal procedure:

1. draw $X \sim \mathcal{N}(0, \Sigma_p)$;
2. draw the residual outcome and quantum output according to $\mathcal{I}(\Phi_0)$;
3. report the classical value $Y = \sqrt{\gamma}X + r$.

Since $\sqrt{\gamma}X \sim \mathcal{N}(0, \gamma\Sigma_p)$, conditioning on r gives the density

$$y \longmapsto \varphi_{0, \gamma\Sigma_p}(y - r).$$

Testing this measure against $a(y)M$ yields exactly (3). The calculation is initially justified for continuous compactly supported a ; a monotone-class argument extends it to bounded measurable a .

6 A classical analogy

Ignore the quantum component and suppose K is a Markov kernel satisfying

$$K(E \mid x + s) = K(E - \sqrt{\gamma}s \mid x).$$

Taking $s = -x$ gives

$$K(E \mid x) = K(E - \sqrt{\gamma}x \mid 0).$$

Thus every such kernel has the form

$$Y = \sqrt{\gamma}X + R,$$

where the residual law of R is independent of X . Lemma C.3 is the classical–quantum version: the residual outcome r may be correlated with a quantum operation, so its law and posterior state must be encoded by an instrument rather than an ordinary probability measure.

7 Dependencies and role in the paper

The construction uses weak-* compactness in much the same way as Lemma C.2, but for a localized family $\Gamma(u_j X)$ rather than for group-averaged channels. It also uses:

- approximate identities in $L^1(\mathbf{H})$;
- covariance and convolution;
- density of finite sums $f(x)X$ in the Bochner L^1 space;
- Radon-measure regularity and trace-norm countable additivity; and
- dominated convergence after testing against $\mathfrak{A}$.

In Proposition C.4, the residual subinstrument isolates all deviations from the ideal classical dilation. Every branch has precisely the displacement and phase covariance needed for Lemma D.4. The Gaussian specialization (3) then separates the final estimate into a classical Gaussian overlap and a quantum thermal-state overlap. This is the source of the product

$$B_{\text{cl}}(\gamma, d)B_{\text{orb}}(\gamma, p).$$

8 Minimal prerequisite checklist

At an intuitive level, the lemma says only that translation covariance forces an additive-noise representation. A rigorous reading additionally requires operator-valued Radon measures, completely positive instruments, weak-* compactness of adjoints, approximate identities, Bochner L^1 spaces, and trace-norm countable additivity. The approximate identity is not an optional complication: it is what makes sense of “the output at a classical point” when the original domain contains only L^1 densities.